

FOOD & BEVERAGE

Predicting Restaurant Rating Categories

Random Forest vs Logistic Regression: A Comparative Analysis of Zomato and Swiggy

Bangalore Restaurant Data | Orange Data Mining

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INTRODUCTION

Why Restaurant Ratings Matter

Every restaurant on Zomato carries a single number, its rating. It quietly decides whether a customer scrolls past or taps in. For restaurant owners, understanding what drives that number is a genuine business question, not just a data science exercise.

This project treats restaurant rating as a classification problem: given a restaurant's cost, popularity, service options and cuisine, can we predict whether it falls into a Low, Medium, or High rating band, and which of two standard algorithms does it better?

THE SCALE

51,717

restaurants scraped from Zomato Bangalore

41,418

clean rows used after preprocessing

3

rating classes: Low / Medium / High

2

algorithms compared head-to-head

OBJECTIVES

What This Analysis Set Out to Do

01

Predict

Classify restaurants into Low / Medium / High rating bands using operational and cost features.

02

Compare

Evaluate Random Forest against Logistic Regression to see which algorithm suits this business problem.

03

Analyze

Identify how restaurant characteristics such as cost, votes and services-related features differ across rating categories.

04

Compare platforms

Compare model performance and rating patterns across Zomato and Swiggy

Dataset Description

Source: Zomato Bangalore Restaurants (Kaggle)

Real scraped listings covering restaurant name, cuisines, cost, online ordering and table booking availability, votes, and business type (delivery, dine-out, cafe, buffet).

Field	Detail
Raw rows	51,717
Rows after cleaning	41,418 (80% retained)
Features used	online_order, book_table, votes, rest_type, listed_in(type), cost_clean, Platform
Target variable	rating_class (Low < 3.5, Medium 3.5–4.0, High > 4.0)
Meta fields kept	name, cuisines

A NOTE ON DATA REALITY

This wasn't a tidy Kaggle-ready CSV. Ratings arrived as text ("4.1/5"), costs contained commas, and a small share of rows carried "NEW" or "-" placeholders instead of a score.

Cleaning was done directly inside Orange's Python Script widget, parsing the rating text, stripping commas from cost, and dropping rows with unusable values, rather than pre-cleaning in Excel. This kept the entire pipeline inside one reproducible Orange workflow.

Dataset Description

Source: Swiggy Bangalore Restaurants (Kaggle)

Real scraped listings covering area, category, cost and restaurant names

Field	Detail
Raw rows	10298
Rows after cleaning	1,499 (14.6% retained)
Features used	Area, Category, cost_clean
Target variable	rating_class (Low < 3.5, Medium 3.5–4.0, High > 4.0)
Meta fields kept	RestaurantName

A NOTE ON DATA REALITY

Unlike Zomato, most of Swiggy's listings simply don't carry a public rating yet. The field showed "NA" for the vast majority of restaurants, likely because they're new or haven't crossed the order-volume threshold Swiggy requires before displaying a score.

After keeping only restaurants with an actual numeric rating, just 1,499 of 10,298 rows (14.6%) remained usable a far steeper drop than Zomato's 80% retention.

A second, more consequential gap: not a single restaurant in this cleaned sample exceeds a 4.0 rating. The target variable collapses to Low and Medium only, with Medium dominating at 82% of rows, meaning this dataset can't test the same three-tier classification task Zomato's data could

Why Classification and Why These Two Models

Rating prediction is naturally a classification problem once scores are binned into Low / Medium / High. Restaurant owners don't need a rating to two decimal places; they need to know which tier they're in and what moves them up.



Logistic Regression

A linear baseline. Assumes each feature (cost, votes, service type) contributes independently and additively to the outcome.

Mechanically: it models the log-odds of each tier as a weighted sum of features — each coefficient reads as “one more unit of cost shifts the odds by X.”

Fast, interpretable, easy to explain, but can't capture interactions between features.



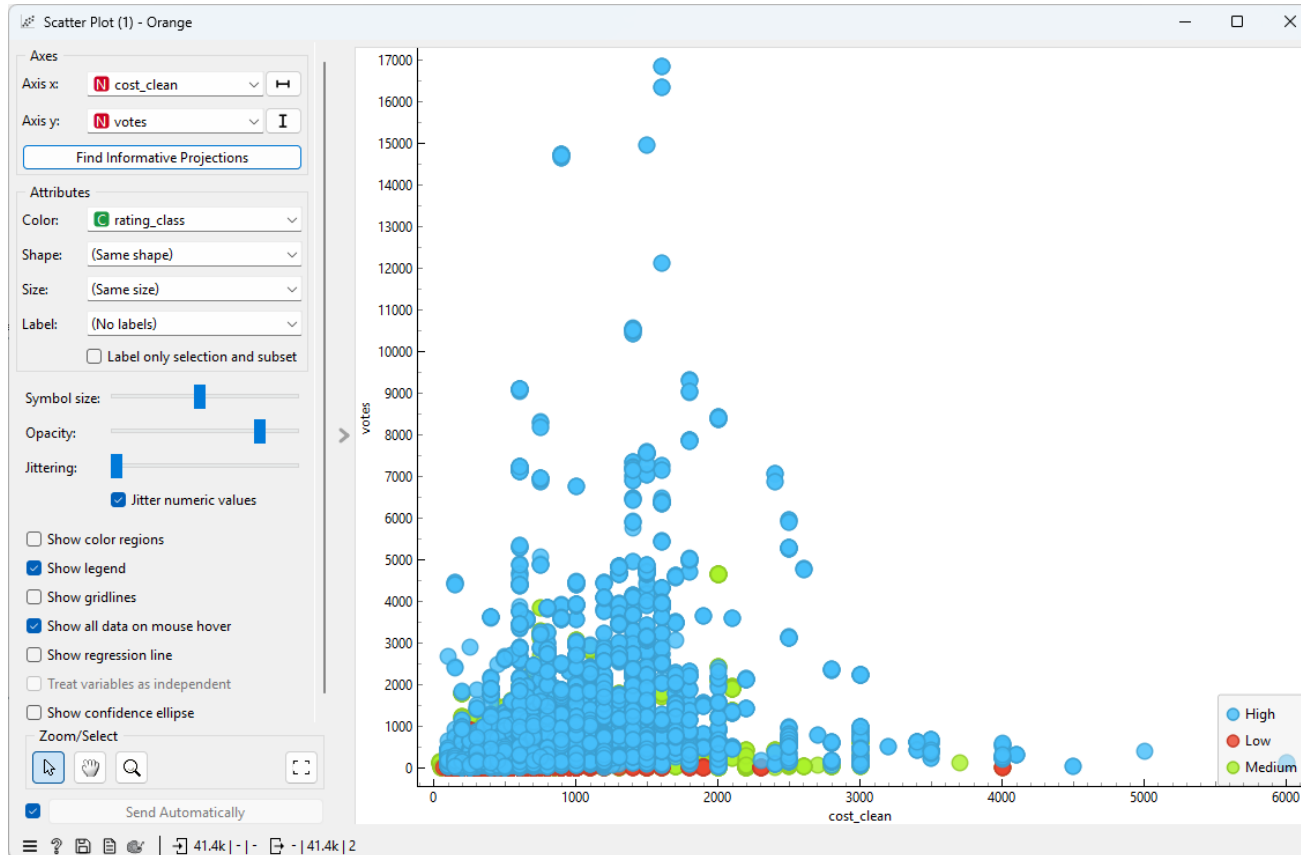
Random Forest

An ensemble of decision trees. Naturally captures non-linear relationships and interactions, e.g. a cheap restaurant with high votes behaves differently than a cheap restaurant with low votes.

Mechanically: it grows many trees on bootstrapped samples; each splits on whichever feature threshold best separates classes at that node, and predictions are majority-voted across all trees.

More flexible, typically stronger on tabular business data like this.

Where the Interaction Shows Up



SCATTER PLOT (ORANGE)

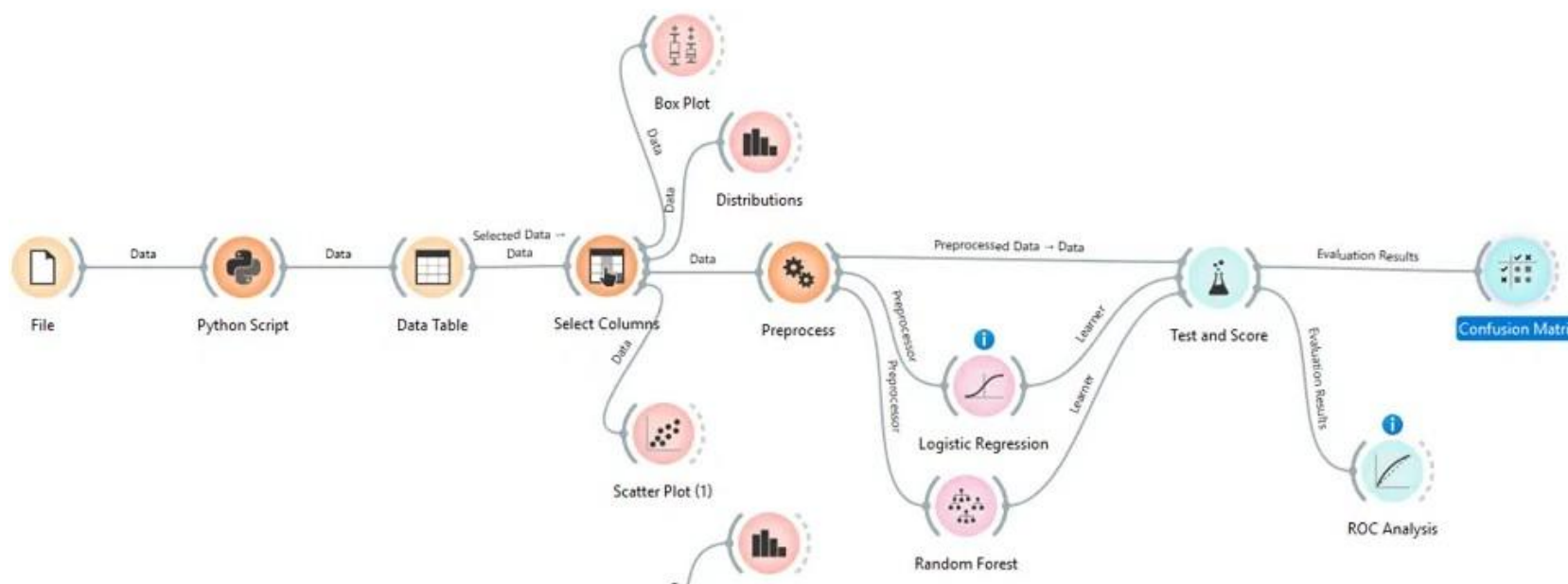
Cost and votes are plotted against each other, coloured by rating_class. Most High-rated restaurants (blue) cluster where both cost and votes are elevated together — not wherever either feature is high alone.

A cheap restaurant with high votes behaves differently from a cheap restaurant with low votes — that is exactly the kind of conditional pattern a linear model can't represent but a tree-based model captures automatically.

Key insight: Logistic Regression would fit one straight-line boundary through this cloud of points. Random Forest splits the space repeatedly on both axes, which is why it separates these tiers far more cleanly.

The Zomato Pipeline

Comparing random forest and logistic regression



1. File → Python Script

Loads zomato.csv; a Python Script widget parses the rating text, strips commas from cost, and builds the rating_class target.

2. Select Columns

Keeps only what modeling needs: cost, votes, service flags and rest type, plus the rating_class target.

3. Preprocess → Learners

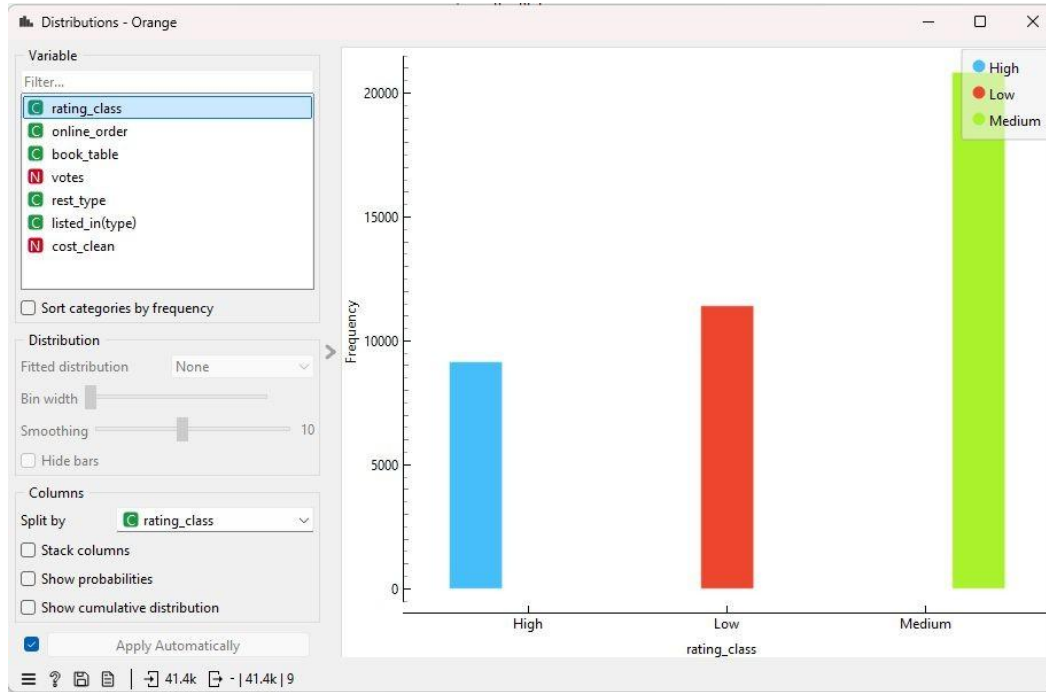
Encodes categorical fields, then feeds identical processed data to both Logistic Regression and Random Forest.

4. Test and Score

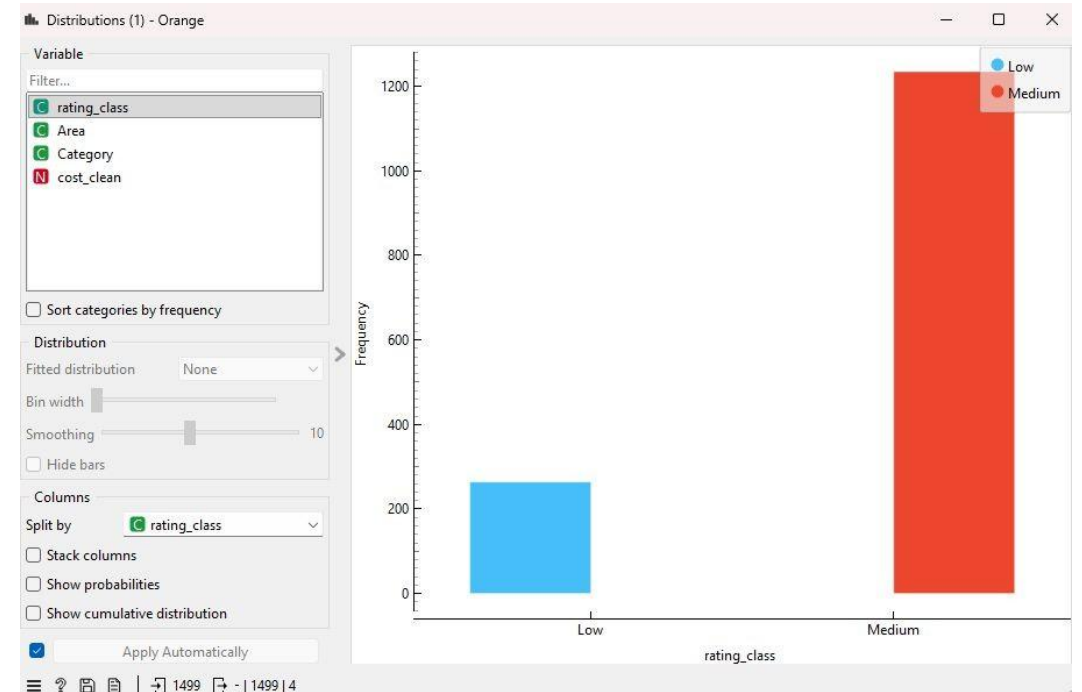
5-fold stratified cross-validation feeds the Confusion Matrix; Box Plot / Distributions run in parallel for exploratory checks.

RESULTS: DATA VISUALISATION

How Are Ratings Distributed?



Zomato



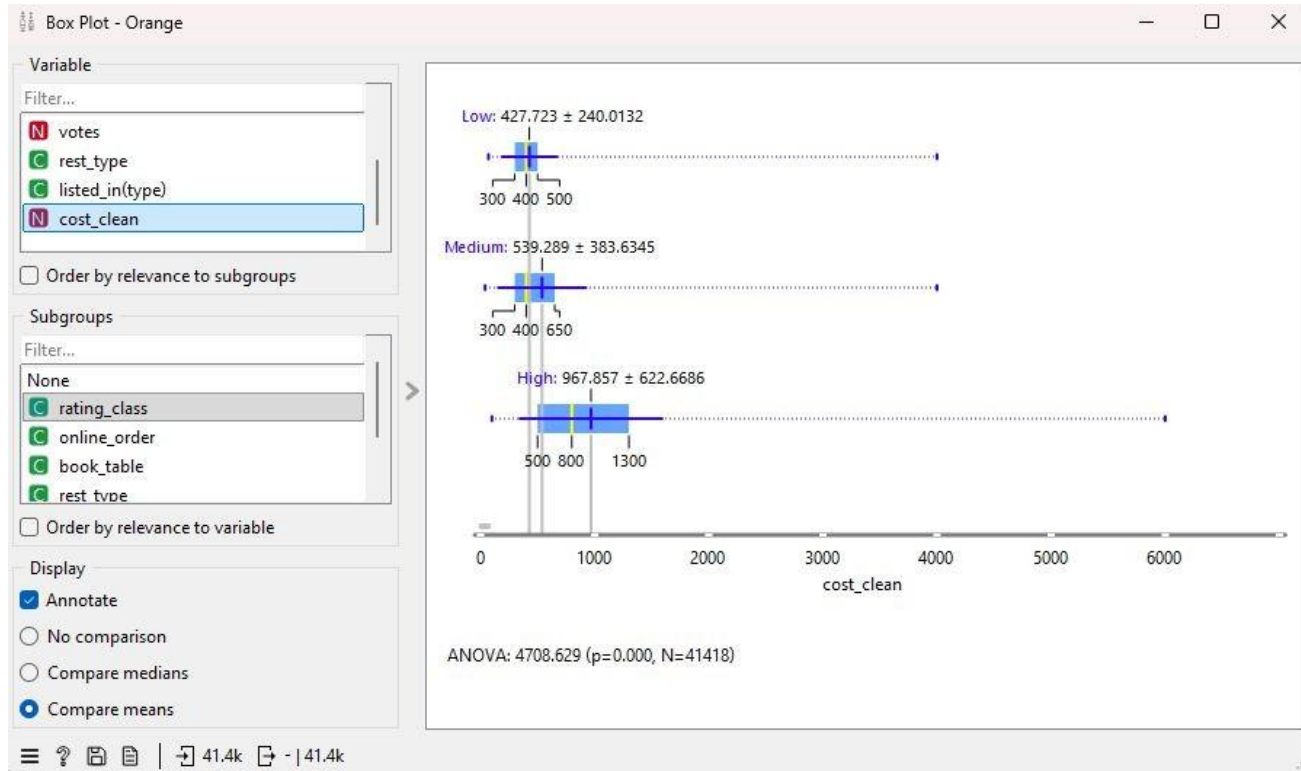
Swiggy

READING THE SPREAD

Zomato shows greater rating variation, allowing cost, votes and restaurant characteristics to provide stronger signals for distinguishing rating classes. Swiggy's limited rating range and heavy concentration in Medium reduce this variation, making classification less informative.

Overall, the platform data structure, not just the algorithm, drives the difference in model performance.

Cost Is the Clearest Visual Signal



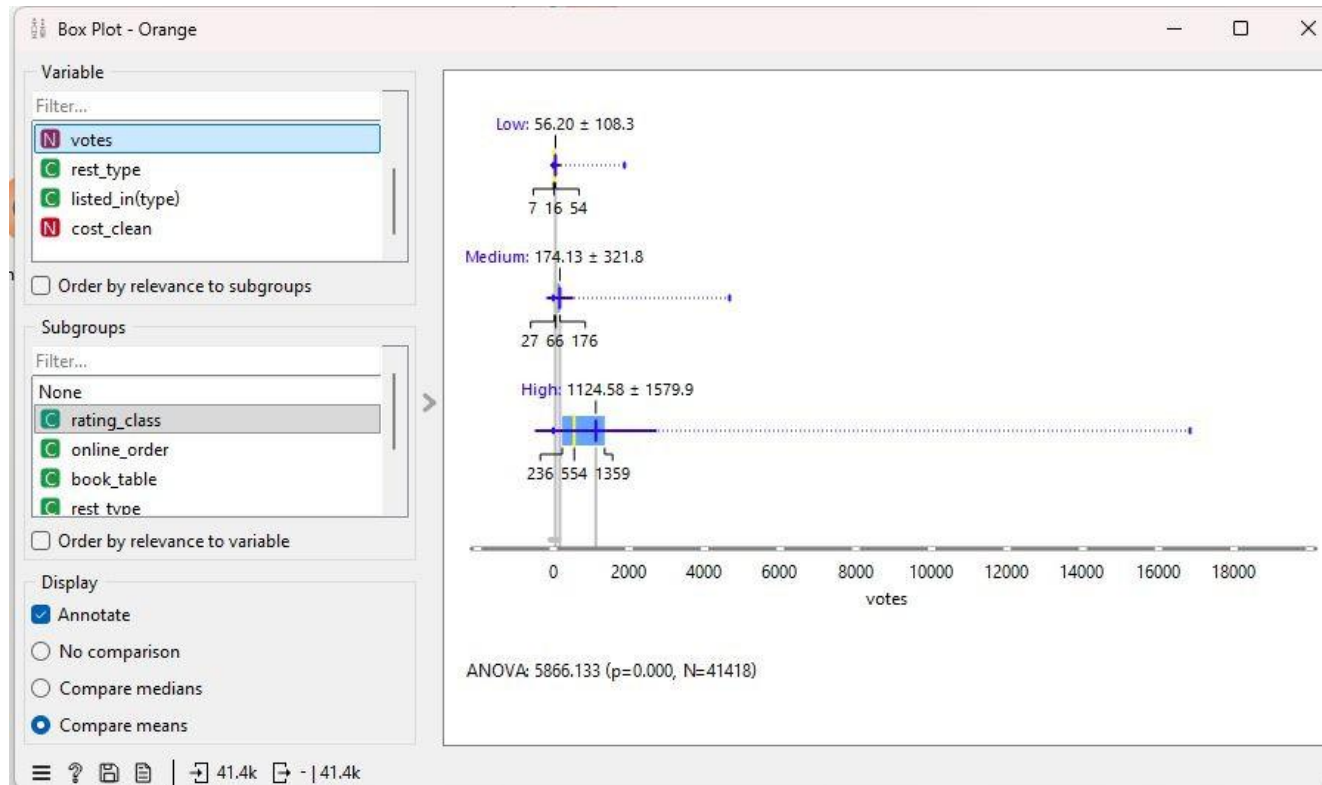
BOX PLOT SUMMARY (ORANGE)

Cost represents the price level associated with a restaurant, which may reflect differences in restaurant positioning, service quality, and customer expectations. The box plot shows that cost tends to increase across the Low, Medium and High rating classes, with greater variation among highly rated restaurants.

A one-way ANOVA confirms that the differences in mean cost across rating classes are statistically significant ($F = 4708.6$, $p < 0.001$). This indicates a strong association between restaurant cost and rating class.

Key insight: Higher-rated restaurants tend to have substantially higher costs, suggesting that price level and perceived restaurant quality are strongly associated in the dataset. However, the analysis shows association rather than causation.

Popularity (Votes) by Rating Class



BOX PLOT SUMMARY (ORANGE)

Votes act as a proxy for customer engagement and restaurant popularity. The box plot shows a clear increase in vote levels from Low to High-rated restaurants, with greater variability among highly rated restaurants.

A one-way ANOVA confirms that the differences in mean votes across rating classes are statistically significant ($F = 5866.1, p < 0.001$). This suggests a strong association between customer engagement and rating class. However, the variation within groups means that higher ratings do not necessarily cause higher popularity.

Key insight: Higher-rated restaurants tend to attract substantially greater customer engagement, making votes an important indicator associated with restaurant rating.

RESULTS: MODEL COMPARISON

Random Forest Wins, Decisively

Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression	0.778	0.629	0.613	0.642	0.629	0.376
Random Forest	0.928	0.823	0.822	0.822	0.823	0.714

5-fold stratified cross-validation • 41,418 rows • evaluated via Orange's Test and Score widget

WHY RANDOM FOREST WINS

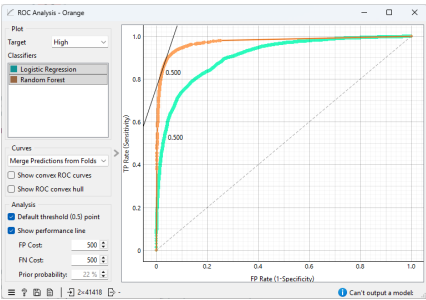
Restaurant rating drivers interact rather than act independently. A cheap restaurant with high votes behaves differently to a cheap restaurant with low votes. Logistic Regression assumes additive, linear effects and cannot capture that; Random Forest's tree ensembles model these interactions naturally, which explains its 19.4-point accuracy lead and near-perfect win probability on paired comparison (1.000 on AUC).

CONFUSION MATRIX — RANDOM FOREST

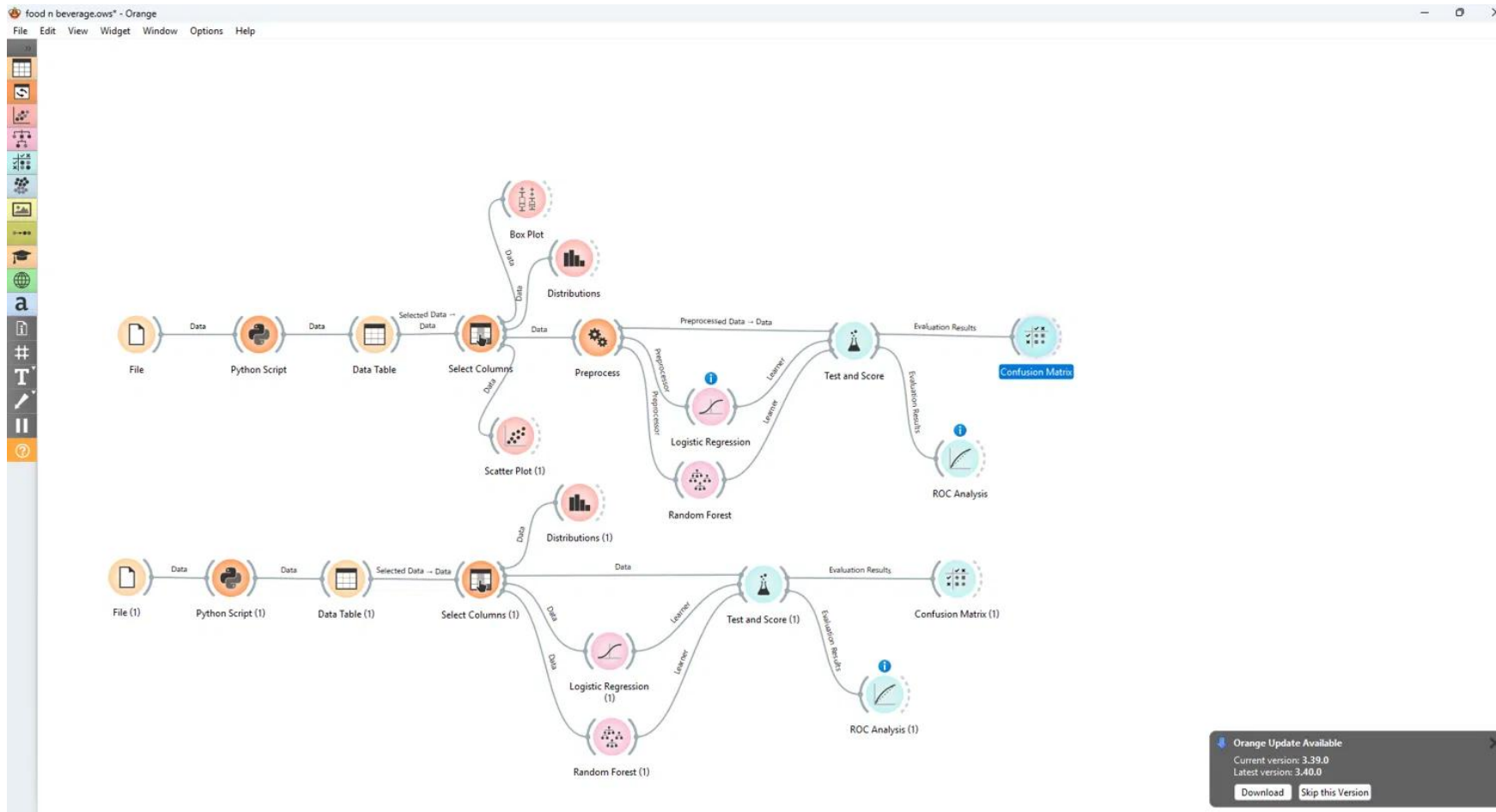
		Predicted			Σ
		High	Low	Medium	
Actual	High	7984	115	1057	9156
	Low	134	8526	2760	11420
	Medium	992	2289	17561	20842
Σ		9110	10930	21378	41418

High and Low are rarely confused with each other. Medium absorbs most misclassifications from both sides, which is expected, since it's the boundary class.

ROC — ZOMATO (RF vs LR)



Comparative Model Pipeline



READING THIS PIPELINE

The top branch is the exact Zomato pipeline from the previous slide, unchanged. The bottom branch mirrors it node-for-node on Swiggy's data — same cleaning logic, same two learners, same 5-fold Test and Score.

Two differences worth flagging: Select Columns (1) keeps Area and Category instead of rest_type and listed_in(type) — Swiggy's raw data simply doesn't carry those fields. And with only 1,499 usable rows, this branch skips a separate Box Plot/Distributions pair of its own.

CROSS-PLATFORM COMPARISON

Zomato vs. Swiggy: Same Algorithm, Different Outcome

ZOMATO — TEST & SCORE

Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression	0.778	0.629	0.613	0.642	0.629	0.376
Random Forest	0.928	0.823	0.822	0.822	0.823	0.714

SWIGGY — TEST & SCORE

Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression (1)	0.600	0.825	0.748	0.856	0.825	0.079
Random Forest (1)	0.537	0.789	0.746	0.720	0.789	0.031

AUC — THE FAIR YARDSTICK

	Zomato	Swiggy
Random Forest	0.928	0.537
Logistic Regr.	0.778	0.600

Zomato's Random Forest separates rating tiers well; on Swiggy both models sit near the coin-flip line — cost and area carry no real signal there.

See the ROC comparison ahead for why Logistic Regression edges out on Swiggy.

ZOMATO — CONFUSION MATRIX

		Predicted			Σ
		High	Low	Medium	
Actual	High	7984	115	1057	9156
	Low	134	8526	2760	11420
	Medium	992	2289	17561	20842
Σ		9110	10930	21378	41418

SWIGGY — CONFUSION MATRIX

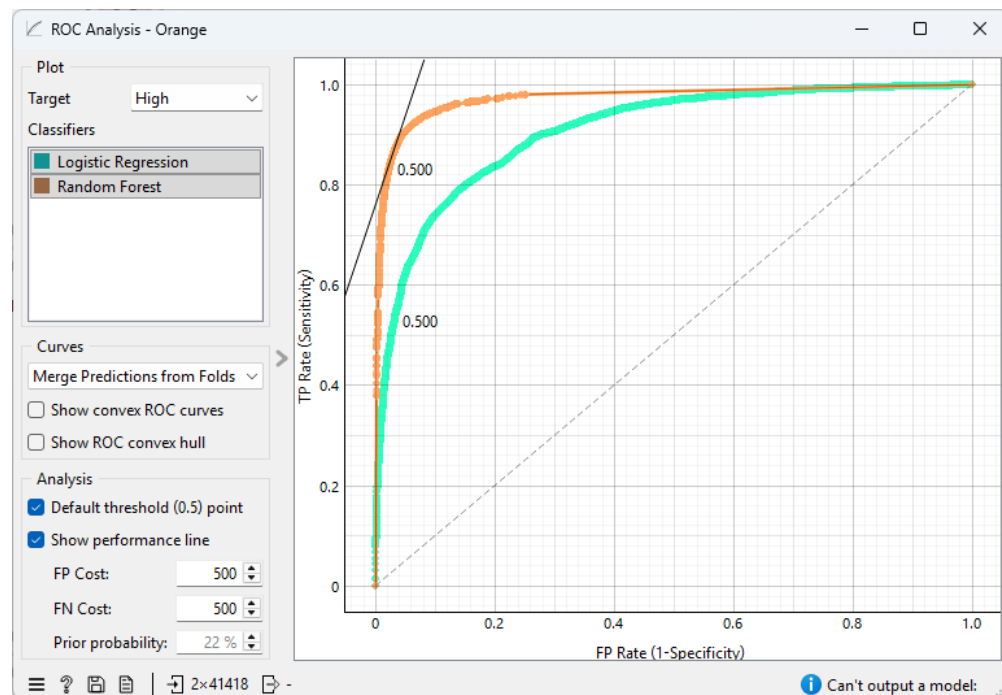
		Predicted		Σ
		Low	Medium	
Actual	Low	2	262	264
	Medium	0	1235	1235
Σ		2	1497	1499

READING THE DIVERGENCE

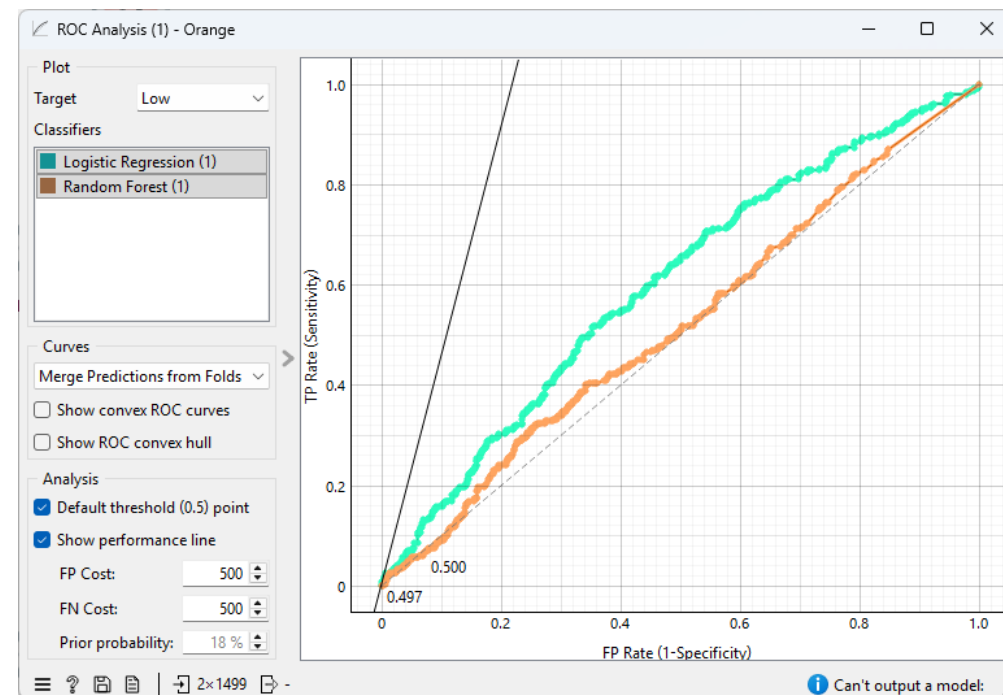
Zomato's target spans all three tiers across 41,418 rows, enough signal for cost and popularity to meaningfully separate them. In this Swiggy sample no restaurant exceeds a 4.0 rating and only 1,499 rows have a published score, so the target collapses to Low vs. Medium with heavy imbalance. CA/accuracy is therefore not comparable across platforms; AUC is used instead as a threshold- and class-count-independent measure. Thus, cost and area explain Zomato ratings well, but not Swiggy's — its rating variation likely hinges on delivery-specific factors (speed, order accuracy) this dataset doesn't capture, consistent with industry framing that Swiggy ratings lean on delivery experience rather than price. For future platform choice, Zomato appears more useful when customers want to differentiate between restaurant quality levels, while Swiggy may be better treated as a platform for delivery-focused decisions rather than relying heavily on its rating alone.

CROSS-PLATFORM COMPARISON

AUC as ROC Curves



Zomato



Swiggy

READING THE CURVES

Zomato's Random Forest (orange) bows sharply toward the top-left, reaching AUC 0.928; Logistic Regression (green) trails at 0.778 but still clears the diagonal comfortably. On Swiggy, both curves hug the diagonal — AUC 0.537 (Random Forest) and 0.600 (Logistic Regression) — meaning neither model beats a coin flip by much.

Logistic Regression's edge on Swiggy is real but small: it leads on AUC, accuracy, precision, recall and MCC — likely because Swiggy's heavy concentration in the Medium class (82% of rows) rewards a model that leans on the dominant class rather than one hunting for interactions this data doesn't support.

Where This Analysis Could Go Next

- Test whether cuisine type interacts with cost: does the cost to rating relationship hold equally for cafes, fine dining, and quick bites?
- Add delivery-time or order-volume data (where available) to see if operational speed rivals cost as a predictor, matching how platforms describe their own ranking logic.
- Extend rating_class to a finer-grained scale (5 bands instead of 3) to see if Random Forest's advantage holds at higher resolution.

REFERENCES

References

1. Zomato Bangalore Restaurants Dataset. Kaggle. (Uploaded by Himanshu Poddar).
2. Swiggy Bangalore Restaurants Dataset. Kaggle.
3. Orange Data Mining Toolbox — University of Ljubljana, Bioinformatics Laboratory. orangedatamining.com

Thank you